



University of Moratuwa

Department of Electronic and Telecommunication

Assignment 01

EN4554 - Deep Learning for Vision

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Part 2: LoRA Fine-tuning a Small LLM for Tool Calling

1. Setup

- **Model:** Qwen2.5-0.5B-Instruct, used through its built-in tool-calling chat template (each call is a JSON object wrapped in `<tool_call>` tags). Greedy decoding, at most 128 new tokens per turn.
- **Tools (4):** `calculator` (arithmetic), `unit_convert` (length, mass, °C/°F), `get_weather` (made-up weather for six Sri Lankan cities) and `lookup_order` (60 made-up orders). The invented values force the model to call the tools instead of answering from memory.
- **Agent loop:** generate → parse any tool call → execute it → append the result as a tool message → repeat, until a plain-text reply or 4 steps.
- **Data:** template-generated, with gold answers computed by running the tools. **Train: 1,200** requests (856 single-call, 118 two-call chains, 226 needing no tool). **Test: 100** requests (68 single-call, 5 chains, 27 no-tool; 77 unique), built from separately written templates with different wording. No test request appears in the training set.
- **LoRA:** $r = 16$, $\alpha = 32$, dropout 0.05 on the q, k, v, o, gate, up and down projections (8.80 M trainable parameters, 1.75 %). Loss on assistant tokens only; system, user and tool-result tokens are masked. 2 epochs, learning rate 2×10^{-4} (cosine, 5 warm-up steps), effective batch 16 (micro-batch 2×8 accumulation, because batch 8 ran out of memory with Qwen's 152k-token vocabulary), fp16 mixed precision, seed 42. 150 steps in 6 min 42 s on a Colab T4; training loss fell from 0.34 to 0.0003.

2. Baseline vs. LoRA (same 100-request test set)

Metric	Definition (denominator)	Baseline	LoRA	Change
Call validity	made ≥ 1 call, all well-formed (tool requests, $n = 73$)	83.6 %	**100.0 %**	+16.4
Call accuracy	right tool(s) and right arguments ($n = 73$)	68.5 %	**90.4 %**	+21.9
Abstention	answered without a tool when none was needed ($n = 27$)	59.3 %	**96.3 %**	+37.0
Task success	final answer contains the gold facts, end to end ($n = 100$)	55.0 %	**92.0 %**	+37.0

The baseline never produced malformed JSON: all of its validity failures were tool requests where it made no call at all.

3. Failure analysis

Category	Base	LoRA
Ignored tool result	15	0
Missed call	14	0
Unnecessary call	11	1
Wrong no-tool answer	7	0
Wrong tool	6	4
Wrong arguments	3	3
Wrong format	0	0
Total failures	**56**	**8**

One category per request; "missed call" includes chains cut short. On the same requests, LoRA **fixed 49** of the 56 baseline failures, **left 7 unfixed** and **introduced 1**. All 8 LoRA failures were inspected (only 8 exist, fewer than the 10 asked for), along with 10 baseline failures.

Baseline errors were decision errors, not format errors. It asked the user for data instead of calling ("should I pack a coat?" → "could you provide the current temperature?"), called tools for general knowledge (`lookup_order(order_id="RAM")` for "What does RAM stand for?"), refused easy questions ("I can only provide details about mathematical operations and weather") and misreported correct results ("the backpack is expected to be back in stock in 6 days"). **LoRA fixed** every ignored result, refusal and wrong-tool error, 10 of the 11 unnecessary calls and 10 of the 14 missed calls.

Not fixed (7): "Take b away from a" has wrong arguments in both models, and LoRA computes $b - a$ (2 cases). The boiling-point question still triggers `calculator("0*100")`, answered as "0 °C" (1). "I'm travelling to *city*, should I pack a coat?" moved from a missed call to the wrong tool, `calculator("1*24")`, answered as "24 hours until you arrive" (4). **Introduced (1):** "Vavuniya weather report, please" now calls `get_weather(city="Lisbon")`.

The remaining LoRA errors share one cause: **over-fitting to the training templates** (loss $\approx 2 \times 10^{-4}$ by epoch 2). The model keys on surface patterns, such as reading "travelling to *city*" as the trained "how many hours until … arrives" chain, and operand order in subtraction, rather than on intent. More varied phrasings, paraphrase augmentation or a single epoch would be the next step. The ignored-result count is also somewhat inflated by a strict answer check that requires both the order status and the item.

4. Would a better prompt alone have been enough?

Only partly: the baseline already produced well-formed calls through Qwen's built-in tool template, so its failures were decisions (whether to call a tool at all in 32 cases, how to report the result in 15), and a stricter system prompt with a few worked examples could plausibly reduce these without any training. However, a 0.5 B model follows long instructions unreliably, and LoRA removed 49 of the 56 failures after under seven minutes of training (we did not run a prompt-only ablation to measure the gap). The errors LoRA left come from template over-fitting, which better training data, not a better prompt, would address.